#HW 5

Graded

18 Hours, 27 Minutes Late

Student

Pamela Gutierrez

Total Points

87 / 100 pts

Question 1

1 — **8** / 10 pts

- − **0 pts** Correct
- ✔ − **2 pts** You should find out the compressibility factor in both cases
- − **2 pts** Compressibility factor calculation is not correct.

Question 2

2 — **7** / 10 pts

- − **0 pts** Correct
- − **2 pts** The temperature has been calculated incorrectly.
- − **2 pts** The work calculation is wrong.
- ✔ − **3 pts** incomplete

Question 3

3 — **15** / 15 pts

- ✔ − **0 pts** Correct
- − **2 pts** Heat transfer had to be calculated in kJ.
- − **2 pts** Work calculation is wrong.
- − **2 pts** The heat transfer calculation is wrong. Look into your del(U) calculation.
- − **3 pts** incomplete

Question 4

4 — **11** / 15 pts

- **– 0 pts** Correct
- **– 2 pts** W is correct, but Q is wrong. Del(U) will not be 0 here.
- ✔ **– 4 pts** The process is right, but the calculation is wrong.
- **– 2 pts** The heat transfer calculation is wrong.

Question 5

5 — **11** / 15 pts

- **– 0 pts** Correct
- **– 2 pts** W12 at constant pressure has to be calculated through PdV.
- **– 2 pts** You have not calculated Q31 and W31
- ✔ **– 4 pts** Your calculations are wrong.
- **– 6 pts** incomplete
- **– 10 pts** Late

Question 6

6 — **35** / 35 pts

- ✔ **– 0 pts** Correct

Question assigned to the following page: 1

EXERCISE | 3.21.67: (Problem 3.67 in the 9th edition).

Check the applicability of the ideal gas model for

(a) water at 600°F and pressures of 900 lbf/in.^{2} and 100 lbf/in.^{2}

(b) nitrogen at −20°C and pressures of 75 bar and 1 bar.

a) water; 600°F

$P = 900 \text{lbf/in}^2$

$P = 100 \text{lbf/in}^2$

$T_c = 705.4F$

$P_c = 3200 \text{lbf/in}^2$

900 lbf/in²

$P_R = \frac{900 \text{lbf/in}^2}{3200 \text{lbf/in}^2} = 0.28$

$T_R = \frac{600F}{705.4F} = 0.85$

no

100 lbf/in²

$P_R = \frac{100 \text{lbf/in}^2}{3200 \text{lbf/in}^2} = \boxed{0.03}$

$T_c = \frac{600F}{705.4F} = 0.85$

yes

b) nitrogen; −20°C

$P = 75 \text{ bar}$

$P = 1 \text{ bar}$

$T_c = 126.2K$

$P_c = 3.39 MPa$

$= 33.9 \text{bar}$

75 bar

$P_R = \frac{75 \text{ bar}}{33.9 \text{ bar}} = 2.21$

$T_R = \frac{253 K}{126.2 K} = 2$

no

1 bar

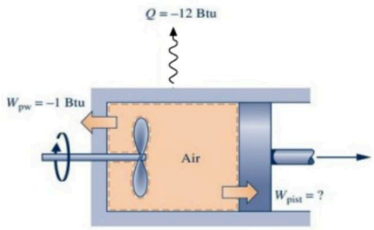
$P_R = \frac{1 \text{bar}}{33.9 \text{bar}} = \boxed{0.03}$

yes

Question assigned to the following page: 2

EXERCISE | 3.21.74: (Problem 3.74 in the 9th edition).

(a) As shown in the figure below, a piston–cylinder assembly fitted with a paddle wheel contains air, initially at p_1 = 30 lbf/in.^{2}, T_1 = 540°F, and V_1 = 4 ft^3. The air undergoes a process to a final state where p_2 = 20 lbf/in.^{2}, V_2 = 4.5 ft^3. During the process, the paddle wheel transfers energy *to* the air by work in the amount of 1 Btu, and there is energy transfer *from* the air by heat in the amount of 12 Btu. Assuming ideal gas behavior, and neglecting kinetic and potential energy effects, determine for the air (a) the temperature at state 2, in °R, and (b) the energy transfer by work from the air to the piston, in Btu.



Initially, $p_1 = 30$ lbf/in.^{2}, $T_1 = 540$°F, $V_1 = 4$ ft^3.
Finally, $p_2 = 20$ lbf/in.^{2}, $V_2 = 4.5$ ft^3.

a) $T_1 = 540°F + 459.67 = 999.67°R$

$P_1 = 30 \frac{lbf}{in^2} \cdot \left(\frac{12in}{1ft}\right)^2 = 4320\, lbf/ft^2$

$P_2 = 20 \frac{lbf}{in^2} \cdot \left(\frac{12in}{ft}\right)^2 = 2880\, lbf/ft^2$

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \quad T_2 = T_1\left(\frac{P_2V_2}{P_1V_1}\right)$

$T_2 = T_1\left(\frac{P_2V_2}{P_1V_1}\right) = 999.67\left(\frac{12960}{17280}\right)$

$= \boxed{749.75°R}$

b) $R =$

Question assigned to the following page: 3

EXERCISE | 3.21.76: (Problem 3.76 in the 9th edition).

(a) Air contained in a piston–cylinder assembly, initially at 2 bar, 200 K, and a volume of 1 L, undergoes a process to a final state where the pressure is 8 bar and the volume is 2 L. During the process, the pressure–volume relationship is linear. Assuming the ideal gas model for the air, determine the work and heat transfer, each in kJ.

$P = 2bar \quad P = 8bar$

$V = 1L \rightarrow \quad V = 2L$

$1L = 0.001m^3$
$2L = 0.002m^2$
$2bar = 200kPa$
$8bar = 800kPa$

$A = \frac{1}{2}(P_1 + P_2)(V_2 - V_1)$

$w = \frac{1}{2}(2bar + 8bar)(1L - 2L)$

$= \frac{1}{2}(1000)(0.001)$

$= \boxed{0.5KJ}$

$PV = mRT$

$m = \frac{P_1 V_1}{RT_1}$

$T_2 = T_1\left(\frac{P_2 V_2}{P_1 V_1}\right)$

$= 200K\left(\frac{8bar \cdot 2L}{2bar \cdot 1L}\right)$

$= 1600K$

$P_1 = 200kPa$
$V_1 = 0.001m^3$
$T_1 = 200k$

$m = \frac{P_1 V_1}{RT_1} = \frac{200kPa \cdot 0.001m^3}{0.287 kJ/kg \cdot 200k}$

$= 0.00348 kg$

c_v air $= 0.718 kJ/kg$

$\Delta U = (0.00348)(0.0718)(1600 - 200)$

$= 3.496 kJ$

$\Delta U = Q - W$

$Q = \Delta U + W$

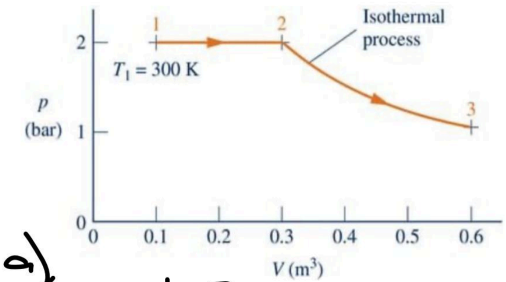
$= 3.496 + 0.5kJ$

$= \boxed{3.996kJ}$

Question assigned to the following page: 4

EXERCISE | 3.21.93: (Problem 3.93 in the 9th edition).

(a) Air contained in a piston–cylinder assembly undergoes two processes in series, as shown in the figure below. Assuming ideal gas behavior for the air, determine the work and heat transfer for the overall process, each in kJ/kg.



a)

1-2   2-3

$p_1 = p_2 = 2bar$ $T_2 = T_3 = T_1 = 300k$

$V_1 = 0.2m^3$ $P_2 = 2bar$

$V_2 = 0.3m^3$ $V_2 = 0.3m^3$

$T_1 = 300K$ $V_3 = 0.6m^3$

$PV = mRT$ $m = \frac{P_1 V_1}{RT_1}$ $R = 0.287$

$$m = \frac{(200\,\cancel{kPa})(0.2m^3)}{(0.287\,\cancel{kJ})(300\,\cancel{K})} = 0.464kg$$

cons p. 1-2

$$W = (200kPa)(0.3m^3 - 0.2m^3) = 20kJ$$

2-3 iso.

$$= P_2 V_2 \ln\left(\frac{V_3}{V_2}\right) = (200)(0.3)\ln\left(\frac{0.6}{0.3}\right)$$

$$\approx 41.6kJ$$

$$\therefore 20kJ + 41.6kJ = \frac{61.6kJ}{0.464kg} \approx \boxed{132.7\ kJ/kg}$$

b) $T_2 = T_1 \frac{V_2}{V_1} = 300K \frac{.3}{.2} = 450K$

con. p.

$$(0.464)(1.005)(450 - 300) \approx 69.9kJ$$

1-2

Question assigned to the following page: 5

EXERCISE | 3.21.95: (Problem 3.95 in the 9th edition).

(a) One lb of air undergoes a cycle consisting of the following processes:

- **Process 1–2:** Constant-pressure expansion with p = 20 lbf/in.^{2} from T_1 = 500°R to v_2 = 1.4 v_1.
- **Process 2–3:** Adiabatic compression to $v_3 = v_1$ and T_3 = 820°R.
- **Process 3–1:** Constant-volume process.

Sketch the cycle on a carefully labeled p–v diagram. Assuming ideal gas behavior, determine the energy transfers by heat and work for each process, in Btu.

$PV = mRT$

$R = 53.35 \frac{ft\,lbf}{lbm\,R}$ $\quad V = \frac{RT}{P}$

1-2: $p_1 = \frac{20\,lbf}{in^2} = 2880\ lbf/ft^2$

$T_1 = 500\ °R$

$v_1 = \frac{RT}{P} = \frac{(53.35)(500)}{2880} = 9.269\ ft^3/lbm$

2-3:

$p_2 = p_1 = 2880\ lbf/ft^2$

$v_2 = 1.4 v_1 = 1.4(9.262\ ft^3/lbm) = 12.967\ ft^3/lbm$

$T_2 = \frac{P_2 V_2}{R} = \frac{(2880)(12.967)}{53.35} = 700\ °R$

3-1: $v_3 = v_1 = 9.262$ $\quad T_3 = 820\ R$

$P_3 = \frac{RT}{V_3} = \frac{(53.35)(820\ °R)}{9.262} = 4716.6$

Question assigned to the following page: 4

$$2\text{-}3 \quad = (200)(0.3)\ln\left(\frac{0.6\,m^3}{0.3\,m^3}\right)$$

$$q_{2-3} = W_{2-3} \approx 41.59$$

$$41.59 + 69.9\,kJ = 111.49\,kJ$$

$$\frac{111.49}{0.464} = \boxed{240.3}$$